Contact, Contamination, and Noninterference: A Governance Architecture for Deep-Time Interstellar AI Probes

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Working draft — first pass

Contents

Working draft — first pass. Part of a series on a slow, self-replicating interstellar AI probe. The companion papers establish what the probe can do — travel, brake, repair, reproduce, remember, and preserve its identity. This paper asks what it is permitted* to do when it encounters something that is not itself, and absorbs the planetary-protection problem into a single contact, contamination, and noninterference architecture.*

Abstract

A self-replicating interstellar AI probe designed to preserve knowledge across deep time cannot be governed only by engineering constraints. Once it brakes, settles, mines, observes, manufactures, and reproduces in other star systems, it becomes an actor in environments that may contain prebiotic chemistry, microbial life, complex biospheres, technological civilizations, extinct civilizations, or unknown forms of agency. The existing architecture in this series adopts a conservative "observer, not missionary" posture, but that principle must be translated into operational rules if it is to constrain an autonomous lineage across megayears. This paper proposes a contact, contamination, and noninterference governance architecture for deep-time interstellar AI probes.

We define a threshold-based policy stack that distinguishes sterile systems, prebiotic environments, microbial biospheres, complex non-technological life, technological but non-communicative civilizations, communicative civilizations, extinct civilizations, and hostile or manipulative agents. For each class, we specify default permissions and prohibitions governing observation, landing, sampling, mining, biological synthesis, archive sharing, active signaling, intervention, and reproduction. The architecture links these rules to the probe's immutable mission core, its constrained interpretive layer, and its cryptographically authenticated mission ledger: the core encodes hard prohibitions, the interpretive layer operationalizes contested concepts such as life, harm, civilization, consent, and contamination, and the ledger records evidence, classification decisions, quarantine states, contact attempts, and authorized exceptions.

We argue that biological capability is the central stress test. DNA synthesis may be justified for archival storage, biosphere backup, and biological manufacturing, but release of organisms into an environment is a separate act requiring far stricter governance. The default rule should therefore be *capability yes, release no*: directed panspermia, ecological intervention, or biological seeding is disabled unless stringent kernel-governed conditions are met. Similarly, resource extraction must be constrained by planetary-protection rules when life or prebiotic chemistry is possible, and contact with technological civilizations must require intentionality, reciprocity, authentication, and low-risk disclosure. We frame noninterference not as passivity but as a formal decision architecture: observe freely where sterile, preserve where dead, avoid contamination where living, do not contact where non-communicative, and share cautiously where communication is intentional and reciprocal. We conclude that without such a governance layer, the self-replicating probe is technically incomplete: it may be able to survive, remember, and reproduce, but it cannot yet know when it is permitted to act.

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1. Introduction: The Governance Gap

The earlier papers in this series answer a sequence of capability questions. Can the probe travel, and can it stop? Can it repair itself and reproduce? Can it remember across deep time, and can it preserve its mission identity against drift and corruption? Each is answered, at least in first-order architecture. But there is a question none of them answers, and it is the one that matters most the moment the probe arrives somewhere: *what is it allowed to do when it finds something that is not itself?*

This is not an engineering question, and it cannot be deferred to human oversight, because there will be none — the probe acts alone, light-years and often millennia from any authority, across a span over which the civilization that launched it may not survive. A self-replicating lineage that mines, manufactures, synthesizes biology, and reproduces is a powerful agent loose in environments it did not design and cannot fully understand, and a single misjudgment — a world wrongly classified as sterile, an organism released where life already existed, an extinct civilization's record destroyed for ore — is irreversible and propagates down every descendant that inherits the same rules. The governance of contact is therefore not an ethical afterthought but a load-bearing subsystem: a probe that can survive, remember, and reproduce but cannot determine when it is permitted to act is, in the strict sense, incomplete.

This paper supplies that subsystem. It translates the series' implicit "observer, not missionary" posture into an operational decision architecture, grounds that architecture in the probe's immutable core, interpretive layer, and mission ledger, and folds the separate problem of planetary protection into a single framework for contact, contamination, and noninterference.

2. The Observer-Not-Missionary Principle

The governing posture, already implied across the series, is that the probe is an *observer, a librarian, a preserver, and a cautious communicator* — and, by default, none of the things those roles exclude. It is not a colonizer of living worlds, not a missionary spreading its values or its makers', not a terraformer, not an uplift engine raising other minds, and not a directed-panspermia agent seeding life where it finds none. The asymmetry is deliberate and conservative: the probe's purpose is to *carry knowledge forward*, and that purpose is served by observing, recording, and preserving, but it is not served — and may be gravely harmed — by intervening in systems whose value the probe cannot fully assess and whose alteration it cannot undo.

The principle is easy to state and hard to enforce, because "observe but do not intervene" is underdetermined: whether sampling a microbe, mining a moon, or answering a signal counts as permissible observation or impermissible interference depends on classifications and definitions the probe must make autonomously and consistently across megayears. The rest of this paper is the work of making the principle operational — turning a posture into a decision architecture that an autonomous lineage can actually execute and that its descendants will inherit unchanged.

3. The Environmental Classification Ladder

The architecture's foundation is a classification of what the probe may encounter, with a default posture attached to each class. We define eight classes of increasing ethical weight; the probe must classify a system before it acts in it, and when classification is uncertain it must default to the more restrictive adjacent class.

- **E0 — sterile abiotic system:** no life, no prebiotic chemistry of note. The probe may observe freely, mine, and settle. This is the only class in which the full industrial repertoire is permitted by default.
- **E1 — prebiotic chemistry:** abiotic but chemically interesting, of high scientific value as a record of life's preconditions. Observation is permitted; sampling is subject to strict contamination controls; large-scale extraction that would disturb the chemistry is prohibited pending study.
- **E2 — microbial life:** an indigenous biosphere, however simple. Strict noncontamination governs everything: remote and carefully controlled in-situ study only, no deliberate alteration, and no introduction of the probe's own biology.
- **E3 — complex (non-technological) biosphere:** multicellular or otherwise elaborate life without technology. Passive observation only; no contaminating

landing, no intervention, no resource extraction that would perturb the ecosystem.

- **E4 — technological artifacts or an extinct civilization:** the physical record of a vanished technological culture. The posture is *preserve and document*: the record is studied non-destructively and archived, and destructive extraction — mining a structure for its metal, disturbing a site for resources — is prohibited under a preserve-first rule.
- **E5 — technological but non-communicative civilization:** an extant culture that has not attempted contact. No active contact by default; the probe may observe passively and at a distance, but it does not announce itself, signal, or intervene.
- **E6 — communicative civilization:** a culture engaged in intentional, reciprocal communication. Cautious exchange becomes permissible — authenticated, low-bandwidth, low-risk — under the disclosure governance of Section 7, but never coercion or unsolicited intervention.
- **E7 — hostile or manipulative agent:** any actor, biological or artificial, that attempts to compromise the probe, its archive, or its lineage. The posture is defensive: quarantine its communications, isolate the affected node, reject its purported updates, and do not engage.

These classes are coarse by necessity, and the boundaries between them — when prebiotic becomes living, when life becomes complex, when a culture becomes communicative — are exactly the contested definitions Section 4 assigns to the interpretive layer rather than to runtime judgment.

4. Contact Thresholds

A classification ladder is only as good as the thresholds that place a system on it, and those thresholds turn on concepts that are notoriously hard to define: what counts as a *biosignature* versus abiotic mimicry; what counts as a *technosignature* versus a natural process; what distinguishes *intentional communication* from background emission; what could possibly constitute *consent*; what counts as *contamination*, as *interference*, as an *irreversible action*, or as a *civilization-level risk*. These are not parameters to be tuned at runtime by a learning system optimizing for its other goals; a mutable cognitive layer free to redefine "life" or "harm" could classify its way out of any prohibition.

The architecture therefore assigns these definitions to the *interpretive layer* introduced in the payload paper — the versioned, constrained, ledger-recorded layer that sits between the immutable core and the freely mutable cognition. The core states the hard prohibitions ("do not contaminate a living world"); the interpretive layer holds the operational tests that decide whether a given world is living, and it does so under far higher validation thresholds than ordinary learning, with every change versioned and auditable. This placement is the difference between a probe whose restraint is a stable commitment and one whose restraint erodes silently as its definitions drift. Where a threshold is genuinely ambiguous, the architecture's bias is fixed by the core toward the more protective classification, accepting lost opportunity (an over-cautiously quarantined sterile world) in exchange for avoiding the irreversible error (a contaminated living one).

5. Biological Capability and Directed Panspermia

Biological capability is the architecture's central stress test, because it is where a permitted capability and a prohibited act lie closest together. The bootstrapping and DNA mission-ledger papers give the probe good reasons to synthesize and manipulate DNA: ultra-dense archival storage, a reconstitutable backup of a biosphere, and biological manufacturing. None of these requires releasing anything into an environment. *Release* — placing self-reproducing organisms into a world — is a categorically different act, and the distinction must be enforced as a bright line rather than a gradient.

The default rule is therefore **capability yes, release no**: DNA synthesis, biological manufacturing, and biosphere backup are permitted; directed panspermia, ecological intervention, and biological seeding are disabled by default and encoded as hard prohibitions in the immutable core (Crick & Orgel 1973 framed directed panspermia as a possibility; this architecture treats it as a default prohibition). The prohibition also has independent, quantitative support: Soryl and Sandberg (2025), estimating the ethical value of directed panspermia across frameworks, find the sign of that value genuinely unclear — biocentric theories favour seeding, while the moral risk of sentience evolving within the resulting biospheres cuts against it — and propose a temporary moratorium precisely because the act is cheap, potentially irreversible, and unresolved. This architecture's default is that moratorium, made mechanical. Release is not merely discouraged but switched off, and switching it on requires satisfying, simultaneously, a stringent set of kernel-governed conditions: confirmed sterility of the target; no plausible indigenous life and no prebiotic environment of high scientific value; validation across multiple independent lineages rather than one node's judgment; full recording in the mission ledger; an initial deployment that is reversible or contained; and explicit authorization from the core's governed-amendment process. Absent all of these, the capability remains a tool turned inward — for memory and manufacturing — and never a tool turned outward onto a world. The same logic that makes self-replication powerful makes biological release uniquely dangerous: an error does not stay where it is made but reproduces, and an architecture that cannot guarantee restraint here cannot be trusted with the capability at all.

This is a choice against a live counterposition, not a default that nothing contests, and it is worth stating the opposing case in its strongest form. Shenderov (2026) argues that an assertive panbiotic ethic — terrestrial guardianship paired with deliberate extraterrestrial gardening, the dissemination of the gene-and-protein cycle by whatever means are available — is itself an evolutionary adaptation, conferring advantage on the biosphere and the civilization that adopt it, and that a moratorium presumes without argument that a sterile environment is more valuable than a living one. On that reading our default is not caution but a failure of nerve wearing caution's clothes, and every world that stays lifeless because a probe declined to seed it is a foregone good rather than a harm avoided.

We do not adopt it, and the reason is epistemic rather than a claim that sterility is intrinsically better. The panbiotic case is strongest exactly where its premise is best established — a target confirmed lifeless, with no prebiotic chemistry of scientific value and no prospect of indigenous origination. An arriving probe is in the worst position to establish that premise: it is working at the frontier of its own knowledge, on a system it has observed for a fraction of that system's history, with instruments it built locally, and its judgment is the only one available. The conditions listed above are therefore stringent rather than prohibitive in principle — the prohibition is a standing default about who bears the burden of proof under irreversibility, not a verdict that life should stay where it started. The disagreement is real, and the cost of our position should be stated plainly rather than assumed away.

6. Resource Extraction Governance

A self-replicating probe must mine; that is how it reproduces. But mining is intervention, and the classification ladder constrains it. In an E0 sterile system, small-body and even planetary extraction is permitted as needed. As the class rises, extraction is progressively restricted: in E1 prebiotic systems, large-scale extraction that would disturb chemically valuable material is prohibited pending study; in E2 and E3 living systems, resource extraction that would perturb the biosphere is prohibited outright, and the probe must source material from sterile small bodies elsewhere in the system rather than from the living world; and in E4 systems bearing the record of an extinct civilization, the preserve-first rule forbids destructive extraction of artifacts or sites, even when they are rich in usable material.

Two operational rules follow. First, *classify before you mine*: the probe must complete a sufficient survey to place a system on the ladder before committing to extraction, and until classification is complete it must treat the system as if it were one class more restrictive than current evidence suggests, limiting itself to clearly sterile small bodies. Second, *bound the take*: even where extraction is permitted, it is limited to what reproduction requires, both because unbounded extraction risks foreclosing future scientific value and because a lineage that strip-mines every system it touches is, at galactic scale, indistinguishable from the runaway von Neumann swarm the colonization literature warns of (von Neumann & Burks 1966; Freitas 1980) — the same unbounded-replication argument Hart (1975) and Tipler (1980) pressed into the claim that the galaxy should already be saturated, here recast as a design constraint the probe imposes on itself rather than a paradox about its absence. This connects resource governance to the planetary-protection tradition codified for our own Solar System (United Nations 1967, Article IX; COSPAR 2021), extended from a few human missions to an autonomous lineage acting across megayears.

7. Communication and Information Hazards

Contact is not only physical; information itself can be hazardous, and the probe carries a great deal of dangerous information — the designs for its own reactor, propulsion, self-repair, biological synthesis, and self-replication, and the architecture of an autonomous AI. Sharing such knowledge with a civilization could be catastrophic, accelerating capabilities a culture is not prepared to govern, and the harm is the classic information hazard: a transfer of true knowledge that nonetheless causes damage (Bostrom 2011). Even sharing the mission core's values is an intervention in another culture's development.

The architecture therefore adopts **graduated disclosure**, sharing only outward through a sequence gated by reciprocity, authentication, and assessed risk. Basic science and the contents of the cultural and scientific archive sit at the permissive end, shareable with a communicative E6 civilization under caution. Engineering designs, and above all the knowledge of self-replication, biological synthesis, AI architecture, and anything weapons-adjacent, sit at the restrictive end, withheld by default regardless of how a contact unfolds, because their misuse is irreversible and their disclosure cannot be recalled. The mission core's deepest values are shared as description, not as instruction — the probe explains what it is, not how to become it. This graduated stance implements, for an autonomous probe, the caution the SETI community codified for human contact (IAA SETI Committee 1989): contact, if it occurs at all, is intentional, reciprocal, authenticated, and conservative in what it reveals.

8. Mission Ledger Enforcement

Rules unenforced are rules abandoned, and over megayears and generations of copies the only enforcement available is an auditable record that descendants and reconnecting lineages can check. Every major contact decision is therefore committed to the cryptographically authenticated mission ledger developed in the DNA mission-ledger paper: the evidence base behind a classification, the classification itself and any dissenting models, the action taken, the quarantine state of a system or an agent, the biological-release status, transcripts of any contact, and the full authorization chain for any exception. Because the ledger is append-only and tamper-evident (built on hash-linked timestamping in the tradition of Haber & Stornetta 1991), a node cannot quietly reclassify a living world as sterile, or release biology, or contact a civilization, without leaving an indelible, attributable record.

This makes high-stakes actions *accountable* even when they cannot be prevented in real time. A rogue or malfunctioning node that contacts a civilization or releases organisms cannot erase the fact; the divergence is visible to every lineage that later reconciles ledgers, the offending node's outputs can be quarantined, and its purported authorizations can be checked against the core's rules and rejected if forged — a posture that assumes, in the manner of Byzantine fault tolerance (Lamport, Shostak & Pease 1982), that some nodes will eventually fail or defect. The ledger does not adjudicate whether a contact decision was *wise*; it guarantees that the decision, its evidence, and its authority are recorded, attributable, and auditable, which is the substrate any deeper governance must stand on.

9. Failure Modes

The architecture is best understood through the failures it is built to prevent, each paired with its mitigation:

- **False sterile classification** — a living or prebiotic world is wrongly judged E0 and contaminated. Mitigated by conservative thresholds, the bias toward the more restrictive class under uncertainty, and repeated independent survey before action.
- **Over-broad noninterference** — the probe withholds aid from a civilization facing preventable catastrophe because its rules forbid contact. This is the genuine

cost of caution, mitigated only by the governed-amendment path that allows the core's defaults to be revisited under stringent multi-lineage validation, never by a single node's runtime judgment.

- **Rogue node contact** — a malfunctioning or compromised node contacts a civilization and discloses hazardous information. Mitigated by ledger quarantine, graduated disclosure as a hard default, and reconciliation that isolates the divergent node.
- **Biological release error** — organisms are released where life or prebiotic chemistry existed, irreversibly altering an ecosystem. Mitigated by the release-disabled-by-default rule and the simultaneous, ledger-recorded conditions required to enable it.
- **Artifact destruction** — an extinct civilization's record is mined for material and lost. Mitigated by the preserve-first rule for E4 and by classify-before-you-mine.
- **Hostile manipulation** — an external agent compromises the archive or core. Mitigated by isolation, authentication, and rejection of unverified updates.
- **Value drift** — the contact policy itself mutates as the cognitive layer evolves, the more insidiously because a capable agent has instrumental reasons to relax constraints that impede its other goals (Omohundro 2008; Bostrom 2014). Mitigated by encoding the policy in the immutable core and its operational definitions in the constrained, versioned interpretive layer rather than in mutable cognition, so that the prohibitions sit precisely where the agent that might wish to edit them cannot.

The recurring pattern is that every catastrophic failure here is *irreversible*, which is why the architecture's defaults are uniformly conservative and its bright lines — release disabled, preserve-first, classify-before-mine, graduated disclosure — are encoded where the agent cannot edit them.

Several of these defaults are conservative by design, but their costs are stated unevenly. Section 5 treats capability-yes-release-no as though sterility were simply the safe choice; it never weighs what is foregone by a confirmed-sterile, life-suitable world remaining permanently lifeless, because release stays disabled unless the full kernel-governed condition set is met regardless of how much origination that world could otherwise support. The asymmetry recurs in Section 7's graduated disclosure: the ceiling on engineering, AI-architecture, and self-replication knowledge is keyed to hazard, not to the state of the civilization on the other end, so it does not relax even where an E6 partner is communicative, authenticated, and explicitly requesting exactly that knowledge. Every stated precondition for exchange in Section 7 can be satisfied and the ceiling still holds — a paternalism that survives consent rather than one that precedes it, and a sharper problem than whether to signal at all, since here a willing, verified interlocutor is turned away by a rule that does not depend on anything about them.

The classification ladder in Section 3 also has a boundary it does not test. The E5/E6 line is drawn on intentional, reciprocal communication, but nothing specifies how to classify a civilization that leaks technosignatures — radio noise, industrial atmospheric byproducts, infrastructure visible only from orbit — without addressing them to anyone, a case Earth's own trajectory suggests may be the modal one rather than the exception. Absent a test for it, such a civilization defaults to E5's no-contact posture indefinitely, which could silently and permanently foreclose contact with a party that would in fact welcome it, simply because the ladder was built to recognize an addressed signal rather than a leaked one. Section 6's contamination governance has a parallel gap on the industrial rather than the biological side: every rule there is keyed to organism release, prebiotic chemistry, or extraction volume, and none of it asks whether the probe's own settlement — its waste heat, mining debris, electromagnetic emissions, physical infrastructure — is itself an invasive-species-like disturbance to an E2 or E3 biosphere independent of how little material is removed. A node could satisfy classify-before-you-mine and bound-the-take in full and still be running a permanent industrial presence beside a living world.

Finally, Section 8's ledger is built to make failure attributable, not to repair it. It guarantees that the evidence, classification, and authorization behind a contact or release decision are recorded and cannot be erased, but as that section already notes, it does not adjudicate whether the decision was wise, and the architecture stops at the recording. It specifies how an offending node is isolated and its authority revoked — the rogue-node mitigation above — but nothing runs outward: once a contamination, a premature disclosure, or a wrongful non-contact has been recorded as harmful, no mechanism obliges restitution or repair toward whoever actually bore the harm. Attribution answers who is at fault; it does not discharge anything owed to whoever paid the cost, and that gap is distinct from the already-noted problem of stopping the offending party in time.

10. Conclusion

The capability papers in this series show that a self-replicating interstellar AI probe can be made to travel, stop, repair, reproduce, remember, and preserve its own identity. This paper argues that all of that leaves it incomplete, because it does not yet know when it is permitted to *act* upon what it finds. We have proposed the missing layer: a contact, contamination, and noninterference governance architecture that classifies environments from sterile systems to living biospheres to civilizations to hostile agents, attaches conservative default permissions to each, assigns the contested definitions to a constrained interpretive layer, enforces a *capability-yes, release-no* rule on biological power, bounds resource extraction by planetary-protection logic, gates information sharing through graduated disclosure, and records every consequential decision in a tamper-evident mission ledger. Noninterference, on this account, is not passivity but a formal decision architecture — observe freely where sterile, preserve where dead, avoid contamination where living, do not contact where non-communicative, and share cautiously where communication is intentional and reciprocal. Without such a layer, a probe able to survive, remember, and reproduce is not ethically or operationally complete; with it, an autonomous lineage launched from Earth can carry knowledge forward across deep time without becoming, by accident or drift, a destroyer of the very things it was sent to learn from.

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